

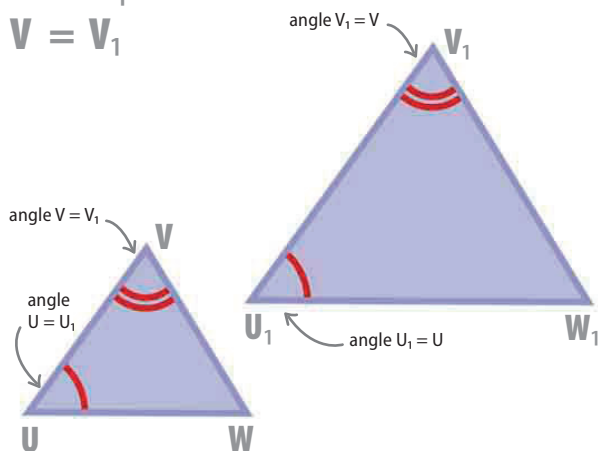
WHEN ARE TWO TRIANGLES SIMILAR?

It is possible to see if two triangles are similar without measuring every angle and every side. This can be done by looking at the following corresponding measurements for both triangles: two angles, all three sides, a pair of sides with an angle between them, or if the triangles are right triangles, the hypotenuse and another side.

Angle, angle AA

When two angles of one triangle are equal to two angles of another triangle then all the corresponding angles are equal in pairs, so the two triangles are similar.

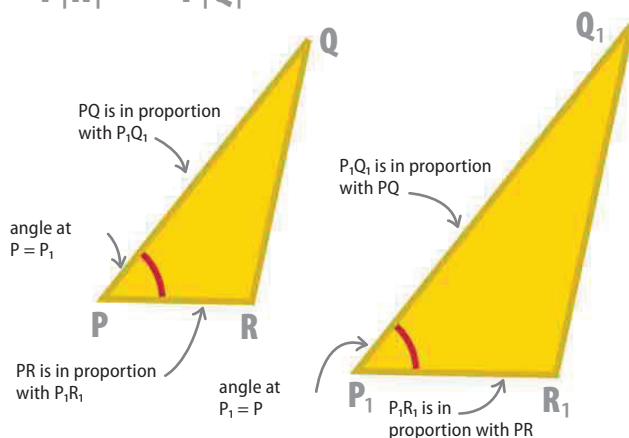
$$\begin{aligned} U &= U_1 \\ V &= V_1 \end{aligned}$$



Side, angle, side (S) A (S)

When two triangles have two pairs of corresponding sides that are in the same ratio and the angles between these two sides are equal, the two triangles are similar.

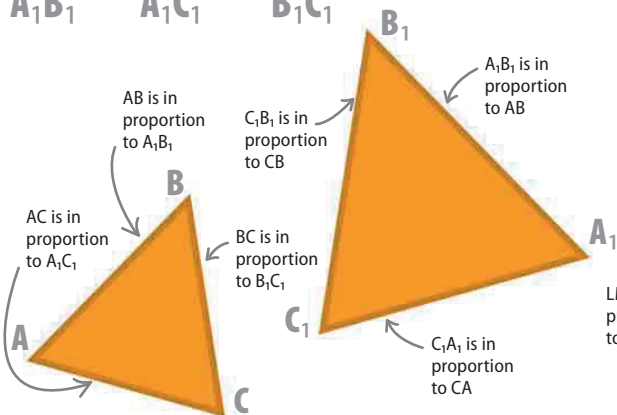
$$\frac{PR}{P_1R_1} = \frac{PQ}{P_1Q_1} \quad \text{and} \quad P = P_1$$



Side, side, side (S) (S) (S)

When two triangles have three pairs of corresponding sides that are in the same ratio, then the two triangles are similar.

$$\frac{AB}{A_1B_1} = \frac{AC}{A_1C_1} = \frac{BC}{B_1C_1}$$



Right-angle, hypotenuse, side R (H) (S)

If the ratio between the hypotenuses of two right triangles is the same as the ratio between another pair of corresponding sides, then the two triangles are similar.

$$\frac{LN}{L_1N_1} = \frac{ML}{M_1L_1} \quad \left(\text{or} \quad \frac{MN}{M_1N_1} \right)$$

